

Anu Varghese

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SUMMARY

Heath Data Scientist & Analytics professional applying Python, SQL, and **Machine Learning** to healthcare and senior-care datasets to drive growth and operational decisions. Experienced in building **HIPAA-aligned** workflows, automated pipelines in BigQuery/Spark, and interactive **KPI dashboards** that support market strategy, pricing, and site-investment decisions. Background includes healthcare outreach **NLP and multi-channel marketing analytics** (Google Ads, Meta Ads) to improve engagement and campaign performance.

TECHNICAL SKILLS

Healthcare Informatics: EHR Data Structures, HIPAA Compliance, FHIR Standards (Interoperability), ICD-10 & CPT Code Analysis, Patient Risk Stratification, Claims Data Processing.

Data Science & ML: Python (Pandas, NumPy, Scikit-learn), R, SQL (Advanced Window Functions), Predictive Modeling, XGBoost, NLP (Text Mining), A/B Testing, Regression Analysis.

Data Engineering: ETL Pipeline Design, Google BigQuery, Snowflake, Redshift, Apache Spark/PySpark, JSON Parsing.

Visualization: Tableau, Power BI, Plotly, Dashboarding, Looker Studio, KPI Development.

Tools: Git, Docker, Jupyter, VS Code, Google Cloud Platform (GCP), AWS, Amazon Bedrock, Snowflake, BigQuery, CRM.

PROFESSIONAL EXPERIENCE

A&R Creative Ventures

Saint Louis, MO, USA | Oct 2025 – Present

Health Data Scientist & Market Analyst

- Built robust predictive models using **Python (Pandas, Scikit-learn)** on **Google Cloud Platform (GCP)** to forecast demand for senior housing, analyzing complex demographic datasets and chronic condition prevalence to identify high-growth market opportunities
- Architected automated data collection frameworks to capture patient needs, establishing pipelines that ingest unstructured qualitative feedback into **BigQuery** to drive **real-time KPI tracking** and evidence-based operational improvements.
- Conducted in-depth competitive analysis of regional healthcare facilities, utilizing **geo-spatial analytics** to optimize market positioning strategies and calibrate service pricing models based on local population health indicators.
- Collaborated with cross-functional executive teams to translate complex model outputs into actionable strategic insights, delivering interactive dashboards that directly supported multi-million dollar investment decisions for new facility sites.

Saint Louis University

Saint Louis, MO, USA | Sep 2024 – Aug 2025

Graduate ML Engineer | Teaching Assistant & Tutor – Health Data Science

- Mentored a cohort of 60+ graduate students in analyzing **Electronic Health Records (EHR)**, guiding them through rigorous **HIPAA-compliant workflows** and data anonymization techniques to ensure patient privacy in research projects.
- Developed and delivered advanced curriculum on **clinical risk prediction**, teaching students how to feature-engineer predictive variables from raw **ICD-10 diagnostic codes** and **CPT procedural billing codes** for healthcare outcomes modeling.
- Led technical workshops on **FHIR interoperability standards**, demonstrating methods to parse JSON healthcare resources using Python and visualize complex patient data relationships within interactive **Tableau** dashboards.

Influnz Media

India | Dec 2020 – Aug 2023

Data Scientist / ML Engineer (Multi-client: Healthcare, EdTech, etc.)

- Designed and deployed a real-time recommendation system for an EdTech engagement (Seedhouse Learning) using **XGBoost + Kafka**, cutting processing latency by **60%** and improving personalization speed.
- Built scalable ML delivery pipelines with **Docker** on AWS EC2, ensuring high availability and stable inference under high concurrency.
- Delivered **NLP sentiment classification** for healthcare outreach programs, increasing patient engagement by 20% via targeted communications.
- Analyzed multi-channel **marketing performance data (Google Ads, Meta Ads)** and built automated reporting pipelines using **Spark + Redshift**, reducing manual reporting by 40% and improving campaign optimization decisions (**CTR/CAC/ROAS**).

EDUCATION

Master's in Health Data Science | Saint Louis University, Saint Louis, MO, USA

Aug 2023 – Aug 2025

Bachelor of Arts in Economics | Kannur University, Kannur, Kerala, India

Jun 2019 – May 2023

PROJECTS

Breast Cancer QA Chatbot | Python, SQL, AI/ML, Gradio

- Developed a **context-aware** healthcare chatbot using **Semantic Search** and **LLMs** that delivers >90% accurate responses for medical education, implementing strict fairness constraints and a real-time compliance dashboard to ensure safety standards and interpretability for patient-facing applications.

Insurance Cost Analysis | R, SQL, BigQuery, Regression Modeling

- Developed a **multi-variable regression model ($R^2=74\%$)** using R and SQL to quantify the financial **impact of risk factors** like BMI and smoking, producing validated reports that projected 18% cost savings and guided insurer strategies for risk adjustment and regulatory alignment.